



Birla Institute of Technology & Science, Pilani

Pilani Campus

Academic Undergraduate Studies Division

FIRST SEMESTER 2026-2027

COURSE HANDOUT (PART-II)

Date: 2nd Aug 2026

In addition to the Part I (The general handout for all courses appended to the timetable), this portion gives further details regarding the course.

Course No. : PHY F214

Course Title : Electromagnetic and Optics Laboratory

Instructor In-charge : Sajal Mukherjee

Instructors Name : Biswanath Layek, D. D. Pant, Madhukar Mishra, Sajal Mukherjee

AIMS AND LEARNING OBJECTIVE:

This lab consists of experiments on electromagnetism and optics. The aim of this lab is to expose students to various topics in the above subject areas by carefully chosen experiments and hence to integrate the theoretical knowledge they acquire in the corresponding theory courses. Students also gain other skills in the process, such as using various scientific equipments, analyzing data, and estimating errors.

TEXTBOOK: Laboratory Manual is available in the lab and will also be uploaded on the Google Classroom. Reference: Relevant reference materials are specified in the lab manuals.

EVALUATION SCHEME:

Component	Duration	Marks (%)	Marks (200)
Day to Day Performance	4 Hrs / Week	50	100
Comprehensive Lab Exam	1.5 Hrs	20	40
Comprehensive Written Exam	1.5 Hrs	30	60

The expectations from students such as attendance, maintenance of lab record book, participation in the lab, etc., will be announced in the first lab.





Laboratory Experiment Plan:

S.N.	Experiment Name	Learning Outcome
1	Brewster Angle & Malus's Law	- Understand polarization of light by transmission through a polarizer - Verify Malus's law of polarization intensity variation with analyzer angle
2	Gaussian Nature of Laser Beam	- Determine the transverse intensity profile (cross-sectional shape) of a laser beam - Measure the beam waist (spot size) and verify its Gaussian distribution
3	Numerical Aperture of Optical Fiber	- Determine the numerical aperture of a multimode optical fiber - Study the effect of fiber bending on the intensity of transmitted light
4	Ultrasonic Diffraction	- Observe diffraction of light by an acoustic grating formed on a liquid surface - Determine the velocity of ultrasonic waves propagating in the liquid medium
5	Michelson Interferometer	- Understand the fundamental principles of interference and optical path difference - Measure the wavelength of a monochromatic source using interference fringes
6	Fresnel's Biprism	- Demonstrate the formation of two virtual coherent sources of light - Determine the wavelength of sodium light using biprism interference fringes



Birla Institute of Technology & Science, Pilani

Pilani Campus

Academic Undergraduate Studies Division

7	Charging and Discharging of a Capacitor	• Study the time dependence of current and voltage in an RC circuit • Determine the time constant ($\tau = RC$) of a given RC network • Estimate the leakage resistance of a practical capacitor
8	Measurement of Dielectric Constant	• Measure the capacitance of a parallel-plate dielectric cell • Determine the dielectric constant of glass and PZT (piezoelectric) materials
9	M–B Curve (Hysteresis Loop)	• Study the magnetization process of a ferromagnetic material • Determine retentivity and coercivity from the magnetic hysteresis loop
10	Measurement of Magneto-resistance	• Investigate the effect of an external magnetic field on the resistivity of materials • Measure the magneto-resistance of a semiconductor crystal
11	Biot–Savart Law	• Study the magnetic field produced by a current-carrying conductor • Experimentally verify the Biot–Savart law of electromagnetism
12	Current Balance	• Measure the force experienced by a current-carrying conductor in a magnetic field • Verify the Lorentz force and the principle of electromagnetic induction



Birla Institute of Technology & Science, Pilani

Pilani Campus

Academic Undergraduate Studies Division

Important Instructions to the students:

1. Day-to-day performance would carry 20 (15 for Performing Experiments, completing results/graphs and calculations, and submitting in stipulated time and 5 for viva) marks for each experiment.
2. We will take the best (N-1) experiments (where N is the total number of experiments performed by a particular Section during the semester) while doing the evaluation at the end of the semester. Total marks for day-to-day performance would be scaled down to **80 marks**.
3. There will be 4 lectures delivered by the Lab instructors which are mandatory to attend. At the end of each lecture, there will be a short quiz of 5 marks based on the topics discussed in the lecture. These **20 marks** will be added to day-to-day performance.
4. Students would perform 1 experiment per turn and upload the calculations and the results on the Google Classroom before the next turn (**6 days**) in the **prescribed format**. **If they fail to do so, 1 mark per day post deadline would be STRICTLY deducted from day-to-day performance for that experiment, and after 3 days post deadline the experiment WILL NOT be evaluated.**
5. Students are expected to read about the allotted experiments from the manuals before coming to the Lab. Viva for each experiment will be conducted on the same day of the experiment.
6. **The Lab Report should be written in the prescribed format only. It should contain the scanned copy of the hand written calculation with your NAME and ID. Similarly the graphs/plots in the report should also contain your NAME and ID, so that they can be authenticated as genuine. In addition to the lab Report, merge the scanned copy of the relevant Observation Sheet and submit it as a single Pdf file only. The name of that Pdf file should be <Your Name_Name of The Experiment.pdf>.**
7. Students are expected to maintain discipline during the lab hours.
8. **As each lab is an evaluative component, mobile phones and any other electronic gadgets are strictly prohibited during lab hours. If found, students will be asked to leave the lab and will be given zero for that experiment.**

Responsibilities of the students:

1. To carry his/her valid identity card issued by the institute.
2. To arrive at the laboratory in time.
3. To get one reading in the observation table verified and attested by the instructors.
4. To understand and strictly follow the safety precautions.

